School Worksheet

Class 12 Mathematics, Chapter 4: Determinants

Maximum marks: **54** **20** questions

Name _____ Date _____ Score _____ / 54

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

8 marks

1 MCQ — 1 mark

If $\begin{vmatrix} x & 2 \\ 18 & x \end{vmatrix} = \begin{vmatrix} 6 & 2 \\ 18 & 6 \end{vmatrix}$, then x is equal to:

(A) 6 **(B)** ± 6 **(C)** -6 **(D)** 0

2 MCQ — 1 mark

If A is a square matrix of order 3 with $|A| = 5$, then $|\text{adj } A|$ is:

(A) 5 **(B)** 25 **(C)** 125 **(D)** $\frac{1}{5}$

3 MCQ — 1 mark

If A is a square matrix of order 3 and $|A| = 4$, then $|2A|$ is:

(A) 8 **(B)** 16 **(C)** 32 **(D)** 64

4 MCQ — 1 mark

The area of the triangle with vertices $(-2, -3)$, $(3, 2)$ and $(-1, -8)$ is:

(A) 15 sq units **(B)** 30 sq units

(C) 20 sq units **(D)** 10 sq units

5 MCQ — 1 mark

The cofactor of the element a_{12} in $\begin{vmatrix} 2 & -3 & 5 \\ 6 & 0 & 4 \\ 1 & 5 & -7 \end{vmatrix}$ is:

(A) 46 **(B)** -46 **(C)** 4 **(D)** -4

6 Assertion-Reason — 1 mark

Assertion (A): If A is a 3×3 matrix with $|A| = 2$, then $|A\,(\text{adj } A)| = 8$. **Reason (R):** For any square matrix A, $A\,(\text{adj } A) = |A|\,I$.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

Q No.	Question	Marks
7	Short answer Find the values of k if the area of the triangle with vertices $(k, 0)$, $(4, 0)$ and $(0, 2)$ is 4 sq units.	2 marks

Level 2

17 marks

8	Short answer Find the adjoint and inverse of $A = \begin{bmatrix} 2 & -1 \\ 4 & 3 \end{bmatrix}$.	2 marks
9	Short answer If $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$, show that $A^2 - 5A + 7I = O$. Hence find A^{-1}.	3 marks
10	Short answer Using determinants, find the equation of the line joining the points $(1, 2)$ and $(3, 6)$.	2 marks
11	Short answer Solve the system of equations using the matrix method: $5x + 2y = 4$, $7x + 3y = 5$.	3 marks
12	Short answer Examine the consistency of the system: $x + 3y = 5$, $2x + 6y = 8$.	2 marks
13	Short answer If $A = \begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix}$, show that $A^{-1} = \frac{1}{19}A$.	2 marks
14	Short answer If $A = \begin{bmatrix} 2 & 3 \\ 1 & -4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 \\ -1 & 3 \end{bmatrix}$, verify that $(AB)^{-1} = B^{-1}A^{-1}$.	3 marks

Level 3

29 marks

15	Long answer A school wants to allocate students into three clubs: Sports, Music and Drama, under the following conditions: (i) the number of students in the Sports club should be equal to the sum of the numbers of students in the Music and Drama clubs; (ii) the number of students in the Music club should be 20 more than half the number of students in the Sports club; (iii) the total number of students allocated to the three clubs is 180. Using the matrix method, find the number of students allocated to each club.	5 marks
16	Long answer Find the product $\begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix}\begin{bmatrix} -2 & 0 & 1 \\ 9 & 2 & -3 \\ 6 & 1 & -2 \end{bmatrix}$ and use it to solve: $x - y + 2z = 1$, $2y - 3z = 1$, $3x - 2y + 4z = 2$.	5 marks
17	Long answer If $A = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 3 & 2 \\ 3 & -3 & -4 \end{bmatrix}$, find A^{-1}. Hence solve: $x + 2y - 3z = -4$, $2x + 3y + 2z = 2$, $3x - 3y - 4z = 11$.	5 marks

Q No.	Question	Marks
18	Long answer The cost of 4 kg onion, 3 kg wheat and 2 kg rice is ₹60. The cost of 2 kg onion, 4 kg wheat and 6 kg rice is ₹90. The cost of 6 kg onion, 2 kg wheat and 3 kg rice is ₹70. Find the cost of each item per kg by the matrix method.	5 marks
19	Long answer If $A = \begin{bmatrix} 1 & 2 & 0 \\ -2 & -1 & -2 \\ 0 & -1 & 1 \end{bmatrix}$, find A^{-1}. Using A^{-1}, solve: $x - 2y = 10$, $2x - y - z = 8$, $-2y + z = 7$.	5 marks
20	Case study **Case Study:** A farmer owns a triangular plot of land whose corners, on a map grid, are $A(2, 3)$, $B(-1, 0)$ and $C(3, -2)$ (1 unit = 10 m). He plans to fix a pole at a point $D(k, 1)$ on the boundary line AB. **(i)** Using determinants, find the area of the plot in square units. — 1 mark **(ii)** Find the value of k so that D lies on line AB. — 1 mark **(iii)** Using determinants, find the equation of line AB. — 2 marks	4 marks

Answer key

Chapter 4: Determinants, School Worksheet. Full step-wise solutions are in the solutions sheet.

Level 1

1	(B) ± 6
2	(B) 25
3	(C) 32
4	(A) 15 sq units
5	(A) 46
6	(A) Both A and R are true, and R is the correct explanation of A
7	$k = 0$ or $k = 8$

Level 2

8	$\text{adj}\, A = \begin{bmatrix} 3 & 1 \\ -4 & 2 \end{bmatrix}$, $A^{-1} = \frac{1}{10}\begin{bmatrix} 3 & 1 \\ -4 & 2 \end{bmatrix}$
9	$A^2 - 5A + 7I = O$ shown; $A^{-1} = \frac{1}{7}\begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}$
10	$y = 2x$
11	$x = 2$, $y = -3$
12	Inconsistent (no solution), since $\lvert A\rvert = 0$ and $(\text{adj}\, A)B \neq O$
13	$A^{-1} = \frac{1}{19}\begin{bmatrix} 2 & 3 \\ 5 & -2 \end{bmatrix} = \frac{1}{19}A$ (shown)
14	$(AB)^{-1} = B^{-1}A^{-1} = \frac{1}{11}\begin{bmatrix} 14 & 5 \\ 5 & 1 \end{bmatrix}$ (verified)

Level 3

15	Sports $= 90$, Music $= 65$, Drama $= 25$ students
16	Product $= I$; $x = 0$, $y = 5$, $z = 3$
17	$A^{-1} = \frac{1}{67}\begin{bmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{bmatrix}$; $x = 3$, $y = -2$, $z = 1$
18	Onion ₹5 per kg, wheat ₹8 per kg, rice ₹8 per kg
19	$A^{-1} = \begin{bmatrix} -3 & -2 & -4 \\ 2 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix}$; $x = 0$, $y = -5$, $z = -3$
20	(i) 9 sq units (ii) $k = 0$ (iii) $x - y + 1 = 0$